

Helix Thready — Database Area (Index)

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Rev	Date	Author	Change
1	2026-07-21	swarm (database)	Initial database pack: ERD, relational + vector DDL, indexing, partitioning, retention/archive, migrations
2	2026-07-21	reviewer (database)	Review pass: registered <code>ATM-DB-033</code> (MinIO signed-URL parity); diagram-consistency fixes across the pack
3	2026-07-22	swarm (database, Pass 3)	Depth pass: shipped runnable migrations <code>0002</code> – <code>0007</code> ; full column-level data dictionary in the relational DDL; per-index rationale matrix; HASH/LIST partition DDL; new <code>constraints-and-integrity.md</code> ; source-confirmed that the pgvector adapter's <code>Search</code> emits no tenant filter (strengthens <code>ATM-DB-013</code>)
4	2026-07-22	critic (database, Pass 4)	Completeness pass: added §8 the consolidated 15-test-type coverage matrix (every mandated type mapped to a concrete DB test bank or explicitly marked delegated/N-A with rationale — closes the CONVENTIONS §6 test-coverage requirement for the data layer); backed the previously-asserted <code>security-7.1</code> “encrypted-yet-searchable” claim with a concrete redacted-embedding spec in <code>schema-vector.sql</code>

This is the canonical entry point for the Helix Thready **Database** area. It specifies the complete, implementation-ready data layer: the entity model, PostgreSQL + pgvector DDL, indexing, partitioning/scaling, retention/archive, and the migration strategy — all grounded in the authoritative research (final request §2.1, §3, §6, §7, §14, Q1–Q45) and the in-house `digital.vasic.database` / `digital.vasic.vectordb` / `digital.vasic.embeddings` modules.

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1. Upstream / Downstream dependencies

Upstream (this area consumes):

- **Architecture** (`../architecture/`) — component boundaries, the relational↔semantic split, event model, and concurrency model that the schema realises.
- **Authoritative research** — `helix_thready_research_request_final.md` (decision matrix, §2.1 data layer, §3 workflow, §6 users, §7 assets, Q1/Q2/Q11/Q12/Q14/Q30/Q40/Q41), `helix_thready_subsystem_gaps_and_improvements.md` (P0/P1/P2 gap register).
- **In-house modules** — `digital.vasic.database` (`pkg/migration.Runner` , `Config` , drivers), `digital.vasic.vectordb` (pgvector `Client`), `digital.vasic.embeddings` (`Dimensions()` per provider), `digital.vasic.background` (claim queue), `digital.vasic.storage` (cold tier), `digital.vasic.security` (column encryption).

Downstream (these areas consume this one):

- **API** (`../api/`) — OpenAPI resources map 1:1 to these entities; `/v1/search` uses the vector reference model; pagination uses the time-partition keys.
 - **Deployment** (`../deployment/`) — runs migrations at deploy, provisions read replicas, backup/DR (RPO 1 h / RTO 4 h), MinIO/S3 cold tier.
 - **Testing** (`../testing/`) — DB test bank: migration apply/rollback on real Postgres, `EXPLAIN` index assertions, search p95 < 500 ms, archive chaos tests.
 - **Development** (`../development/`) — the `ATM-DB-*` workable items tracked below.
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2. Documents in this area

Document	Purpose
<code>erd.md</code>	Full entity-relationship model: master ERD + 5 domain ERDs (Mermaid + prose) + entity dictionary + referential-integrity strategy
<code>schema-relational.sql</code>	Production PostgreSQL DDL for every entity (extensions, tables, constraints, triggers, partition parents)
<code>schema-vector.sql</code>	pgvector DDL: collections referencing relational PKs, cosine ANN indexes, dimension policy
<code>indexing.md</code>	Index catalogue (btree/GIN/trigram/FTS/partial/ANN), query-path map, SLO tuning, hot claim index
<code>partitioning.md</code>	Time-partitioned firehose tables, read replicas, maintenance job, pooling & pgvector co-location tuning
<code>retention-archive.md</code>	Keep-indefinitely + per-account overrides, archive lifecycle, GDPR-aware erasure/export
<code>migration-strategy.md</code>	<code>migration.Runner</code> contract, expand-contract, rollback, advisory lock, verified caveats, roadmap
<code>constraints-and-integrity.md</code>	Enforcement-layer model, CHECK-domain catalogue, FK on-delete matrix, append-only audit + soft-ref validation triggers (closes <code>ATM-DB-004</code>)
<code>migrations/0001_init.sql</code>	Runnable migration (Up/Down): tenancy + ingestion + processing core
<code>migrations/0002_classification.sql</code>	Runnable migration: hashtags, categories, join tables
<code>migrations/0003_assets.sql</code>	Runnable migration: skills, skill_runs, generated_artifacts, assets, asset_links
<code>migrations/0004_billing.sql</code>	Runnable migration: plans, subscriptions, usage_records, invoices
<code>migrations/0005_events_audit.sql</code>	Runnable migration: events (partitioned), event_subscriptions, audit_log (partitioned), archived_partitions
<code>migrations/0006_vector_collections.sql</code>	Runnable migration: <code>vectordb_*</code> collection tables (adapter-parity DDL)
<code>migrations/0007_secondary_indexes.sql</code>	Runnable migration (non-transactional path): all secondary + FTS + vector ANN indexes
<code>diagrams/</code>	Mermaid <code>.mmd</code> sources (siblings of every embedded diagram)

Rendered PNG/SVG exported via Docs Chain (§11.4.65). Every diagram in this area has a sibling `.mmd` in `diagrams/` and an immediate multi-paragraph prose explanation per `CONVENTIONS.md` §4.

3. Decision snapshot (from the matrix)

Concern	Decision	Provenance
Relational DB	<code>digital.vasic.database</code> — SQLite dev / PostgreSQL prod; <code>pkg/migration.Runner</code>	<code>[IN-HOUSE: database]</code>
Vector DB	<code>digital.vasic.vectordb</code> pgvector backend, cosine <code><=></code> ; Qdrant swap by config	<code>[IN-HOUSE: vectordb]</code> Q1
Embeddings	<code>digital.vasic.embeddings</code> → HelixLLM <code>/v1/embeddings</code> ; <code>voyage-code-3</code> (1024) / <code>jina-embeddings-v2-base-code</code> (768)	<code>[IN-HOUSE]</code> Q15
Scale	Large / multi-tenant; 10k+ posts/day, 50 TB+ assets → partition + replicas from day one	<code>[OPERATOR]</code> Q2
Retention	Keep indefinitely + per-account overrides	<code>[OPERATOR]</code> Q12
Billing	Subscription + metered from day one	<code>[OPERATOR]</code> Q11
SLO	Search < 500 ms, API p95 < 150 ms	<code>[OPERATOR]</code> Q14
Backup/DR	Daily full + hourly DB incrementals; RPO ≈ 1 h, RTO ≈ 4 h	<code>[OPERATOR]</code> Q41/Q45
Migrations	<code>pkg/migration.Runner</code> (up/down), expand-contract, tested rollback	<code>[IN-HOUSE]</code> Q30

4. Provenance & verification summary

Following the anti-bluff quality bar (`CONVENTIONS.md` §7), claims are tagged VERIFIED (read at source) vs ASSUMPTION.

VERIFIED (read at module source under `/home/milos/Factory/projects/tools_and_research/helix_code/submodules/`):

- `migration.Migration{Version,Name,Up,Down}`, `NewRunner(db, "schema_migrations")`, `Init/Applied/Apply/Rollback/RollbackWith`; each migration applied in one transaction; `Rollback` is inert and instructs `RollbackWith` (`database/pkg/migration/migration.go`).
- The runner's tracking-table writes use `?` placeholders; the pgx transport (`postgres.Client.Exec` / `pgTx.Exec`) does **not** rewrite them — a real Postgres-path caveat (`ATM-DB-001`, see `migration-strategy.md` §8).
- pgvector `Client.CreateCollection` creates the table with `id TEXT PK, embedding vector(N), metadata JSONB, created_at, updated_at` and **no ANN index**; `Search` uses cosine `<=>` returning `score = 1 - distance` (`vector_db/pkg/pgvector/client.go`). ANN indexes are therefore owned by our migrations.
- **(Pass 3, source-confirmed)** `Client.Search` emits `SELECT id, ... , embedding <=> $1 ORDER BY distance LIMIT $2` with **no WHERE clause** — `SearchQuery.Filter` is accepted by `Validate()` but **never used** in the adapter SQL, and only `id + score` are returned. So tenant isolation cannot be delegated to the shipped adapter; it is app-side over-fetch, per-tenant collections, or a Qdrant swap (`ATM-DB-013`, now verified not assumed).
- **(Pass 3)** `DistanceOperator(metric)` maps `cosine<=>` / `dot_product<#>` / `euclidean<->`; `CollectionConfig.Validate` rejects `Dimension < 1`; `Upsert` serialises all metadata values as JSON **strings** (naive encoder) — the tenant filter is string-typed (`vector_db/pkg/pgvector/client.go`, `pkg/client/client.go`).
- `database.Config` exposes `Driver` + pooling knobs; `digital.vasic.database` has **no** partition/retention/archive package (`database/pkg/` set inspected).
- `embeddings` providers expose `Dimensions()` (jina v2 = 768; voyage-code-3 = 1024) — the dimension is model-driven, not hardcoded.

ASSUMPTION / DEFAULT (adjustable, flagged inline): monthly partition granularity; HNSW `m=16` / `ef_construction=64` / `ef_search=40` starting points; retry defaults (5, 2 s, ×2, 5 min cap); audit retention 1 year; plan/limit shapes; the `-- +thready Up/Down` migration file format.

5. Gaps addressed

Every database-relevant item from `helix_thready_subsystem_gaps_and_improvements.md` is addressed with a design plan or a tracked workable item.

Gap-register item (priority)	Addressed in	How
[GAP: <code>vectordb-3.1</code>] pgvector-only; Qdrant unverified; ANN tuning for < 500 ms (P1)	<code>schema-vector.sql</code> , <code>indexing.md</code> §5	ANN indexes owned by migrations (adapter creates none); HNSW tuning table; Qdrant swap behind <code>VectorStore</code> interface; benchmark <code>ATM-DB-012</code>
[GAP: <code>database-3.2</code>] no partitioning/retention helpers; pooling + pgvector co-location tuning (P1)	<code>partitioning.md</code> , <code>retention-archive.md</code>	Native RANGE partitions + <code>pkg/partition</code> helper proposal; pooling & co-location §7; retention resolution + archive job
[GAP: <code>HelixLLM-1</code>] default <code>HashEmbedder</code> is non-semantic; RAG 768 hardcoded (P0)	<code>schema-vector.sql</code> dimension policy	Dimension discovered via <code>Dimensions()</code> (no hardcoded); explicit “set <code>HELIX_EMBEDDING_PROVIDER=llama</code> ”, never <code>HashEmbedder</code> ” warning
[GAP: <code>helix_skills-4.1</code>] no execution engine (P0)	<code>erd.md</code> §5	<code>skills</code> mirror + <code>skill_runs</code> execution ledger backing the BUILD-NEW dispatch engine
[GAP: <code>auth-7.2</code>] HMAC→RS256/EdDSA + RBAC (P1)	<code>erd.md</code> §3, <code>schema-relational.sql</code>	Full RBAC tables (roles/permissions/role_permissions/memberships) back the User

Gap-register item (priority)	Addressed in	How
		Service; signing handled in API/ security area
[GAP: session_orchestrator-2.9] design-only claim registry (P1)	indexing.md §6, migration-strategy.md §7	Postgres <code>FOR UPDATE SKIP LOCKED</code> partial-index claim + advisory lock replace the unimplemented module for Thready's per-post claim
[GAP: security-7.1] encrypted-yet-searchable sensitive data (P2)	schema-vector.sql (sensitive-content spec), erd.md §4/§5, retention-archive.md §6	Sealed <code>bytea</code> columns <code>(session_enc , access_hash_enc , totp_secret_enc);</code> <code>sensitivity / is_encrypted</code> on assets; concrete redacted/ tokenized-embedding mechanism now specified in <code>schema-vector.sql</code> (redact via <code>security/pkg/pii</code> → embed only the non-identifying form; typed-not-valued metadata; ids-only search + RBAC re-check on hydrate); exact-recall searchable-encryption

Gap-register item (priority)	Addressed in	How
		explicitly out-of-MVP-scope

6. Open items

Tracked as `ATM-DB-*` workable items; none is papered over.

ID	[OPEN: ...]	Summary	Plan
ATM-DB-001	migration-runner-pg-placeholders	Runner tracking writes use <code>?</code> ; pgx needs <code>\$n</code>	Dialect decorator or upstream fix; guarded by a real-Postgres apply/rollback test (migration-strategy.md §8.1)
ATM-DB-002	migration-concurrently	<code>CREATE INDEX CONCURRENTLY</code> can't run in runner tx	Non-transactional path for <code>0007_secondary_indexes</code>
ATM-DB-004	db-partition-fk	App- vs DB-enforced FK into partitioned tables	Decide per env; optional composite FK for <code>processing_state</code>
ATM-DB-011	fts-multilang	Per-language FTS (en/ru/sr-Cyrl)	Lang-driven expression index; semantic search is primary meanwhile
ATM-DB-012	(benchmark)	pgvector vs Qdrant against 500 ms SLO	Scaling/benchmark test bank
ATM-DB-013	vector-tenant-isolation	Source-confirmed: the pgvector adapter's <code>Search</code> emits no <code>WHERE / Filter</code> ; metadata post-filter after ANN can under-fill topK for small tenants sharing a large index	Over-fetch (topK×4) + GIN metadata filter (issued by our own SQL, not <code>Client.Search</code>); per-large-tenant collections; Qdrant payload-filter fallback (schema-vector.sql header, indexing.md §5)
ATM-DB-031	keep-archived-vectors-searchable	Per-account flag to retain vectors of archived content	Retention policy flag
ATM-DB-032	gdpr-cold-erasure	Erasure/anonymisation of cold-tier dumps	Depends on archive format choice
ATM-DB-033	minio-signed-url-parity	<code>storage</code> signed URLs are CloudFront/AWS-specific; verify MinIO parity	Storage/deployment concern (schema stores opaque <code>storage_key</code>);

ID	[OPEN: ...]	Summary	Plan
			retention-archive.md §5
ATM-DB-021	account sub-partitioning	HASH sub-partition for whale tenants	Not MVP; escalation path

7. Entity coverage checklist

All entities mandated by the area scope are modelled (see [erd.md §8](#)):

[messengers](#) ✓ · [accounts](#) ✓ · [channels](#) / [groups](#) ✓ · [posts](#) ✓ · [threads](#) ✓ · [replies](#) ✓ · [hashtags](#) ✓ · [categories](#) ✓ · [assets](#) ✓ · [asset_links](#) ✓ · [processing_state](#) ✓ · [skills](#) ✓ · [users](#) ✓ · [roles](#) ✓ · [permissions](#) ✓ · [memberships](#) ✓ · [events](#) ✓ · [subscriptions](#) ✓ · [billing/metering](#) ([plans](#) , [usage_records](#) , [invoices](#)) ✓ · [audit_log](#) ✓. Supporting: [messenger_accounts](#) , [post_hashtags](#) , [reply_hashtags](#) , [post_categories](#) , [hashtag_categories](#) , [skill_runs](#) , [generated_artifacts](#) , [event_subscriptions](#) , [archived_partitions](#) , and the four [vectordb_*](#) collections ✓.

8. Database test-type coverage (15 mandated types)

[CONVENTIONS.md §6](#) requires the pack to cover the **15 mandated test types** [\[CONSTITUTION §11.4.27\]](#) (final request §9.1): *unit, integration, e2e, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA*. The individual docs ship the RED-first skeletons; this matrix is the **single auditable map** of every type to where the data layer exercises it. Per the anti-bluff rule, mocks/stubs are allowed **only in unit tests** — every other row runs against a **real** Postgres 16 + pgvector container (SKIP-OK if no engine, [CONST-035](#)). Types with no data-layer surface are marked **N-A (delegated)** with the owning area and the DB-side facet they feed — they are *not* silently claimed as covered.

#	Test type	Data-layer coverage	Where
1	Unit	Migration <code>splitMarkers</code> loader parse; retention <code>effective_retention</code> resolution; <code>asset_links_one_subject</code> CHECK arithmetic; CHECK-domain widening logic. Mocks allowed here only.	migration-strategy.md §3, retention-archive.md §2
2	Integration	Full migration <code>Apply</code> / <code>RollbackWith</code> on real Postgres and SQLite; pgvector <code>Upsert</code> + <code>Search</code> adapter contract; FK cascade + soft-ref <code>CONSTRAINT TRIGGER</code> accept/reject.	migration-strategy.md §10, constraints-and-integrity.md §8
3	e2e	Ingest → classify → claim → embed → <code>/v1/search</code> hydrate over a live DB; asserts vectors reference rows and search returns ids that hydrate. DB supplies deterministic seed + partition fixtures; the full flow lives in <code>../testing/</code> .	erd.md §7
4	Full-automation	Seeded multi-month corpus replayed through the whole pipeline against a real DB, unattended; partition create-ahead + age-out run on the maintenance cadence. DB supplies the reproducible seed + <code>MonthlyRange</code> job.	partitioning.md §5
5	Security	<code>audit_log</code> append-only trigger rejects UPDATE/DELETE (defence-in-depth over least-privilege grants); GDPR erasure reaches relational + vector + cold-tier; sealed <code>bytea</code> (<code>session_enc</code> / <code>access_hash_enc</code> / <code>totp_secret_enc</code>) never logged; tenant-isolation over-fetch filter correctness.	constraints-and-integrity.md §6, retention-archive.md §6, schema-vector.sql
6	DDoS	Data-layer facet: a <code>post.received</code> claim storm processes exactly-once via <code>FOR UPDATE SKIP LOCKED</code> (no thundering-herd double-work); bounded PgBouncer/pgx pool caps connection-exhaustion blast radius. N-A (delegated) for network-edge DDoS → <code>../api/</code> / <code>../deployment/</code> .	indexing.md §6, partitioning.md §7
7	Scaling	≥10k posts/day ingestion with search p95 < 500 ms concurrently ; month-boundary inserts land in the right partition and prune in <code>EXPLAIN</code> ; ANN over a seeded 1e6-vector fixture.	partitioning.md §9, indexing.md §8
8	Chaos	Crash mid-archive never drops un-copied data (archive-before-drop invariant); maintenance-job-skipped → rows fall into <code>*_default</code> and the alert fires; strongly-consistent reads never routed to a lagging replica.	retention-archive.md §4/§8, partitioning.md §3/§9
9	Stress	Sustained firehose insert with per-partition aggressive autovacuum (<code>scale_factor=0.02</code>) keeping bloat bounded; the partial claim index stays O(1)-ish under a deep pending backlog.	partitioning.md §7, indexing.md §6

#	Test type	Data-layer coverage	Where
10	Performance	Read SLO p95 < 150 ms: <code>EXPLAIN (ANALYZE, BUFFERS)</code> asserts each hot path is Index/Index-Only Scan, never Seq Scan; the claim query uses <code>idx_processing_claimable</code> .	indexing.md §8
11	Benchmarking	pgvector vs Qdrant against the 500 ms search SLO (<code>ATM-DB-012</code>); HNSW <code>ef_search</code> recall/latency curve; write-amplification budget per firehose insert.	indexing.md §5/ §8
12	UI	N-A (delegated) — the data layer has no UI surface. DB owns no <code>.ds-*</code> /screen assets → <code>../design/</code> , client areas.	—
13	UX	N-A (delegated) — no direct UX surface; the DB's Aggressive latency SLOs (Q14: search < 500 ms, API p95 < 150 ms) are the budget UX tests spend against → client/API areas.	index §3
14	Challenges	<code>vasic-digital/challenges</code> scenario bank — DB scenarios: retention-override resolution (channel > account > global), partition age-out + re-attach, tenant-isolation ANN under-fill (<code>ATM-DB-013</code>), CHECK-domain expand widening.	register §9.3, <code>constraints-and-integrity.md</code> §2
15	HelixQA	<code>HelixDevelopment/helix_qa</code> YAML banks with mandatory runtime evidence — migration apply/rollback transcripts + <code>EXPLAIN</code> output + partition-listing captured as evidence artifacts (the org anti-bluff rule; a green SQLite-only run does not satisfy the Postgres-path caveats <code>ATM-DB-001/002</code>).	migration-strategy.md §8/ §10

Anti-bluff note. Rows 6/12/13 are honestly scoped: DDoS at the network edge, and all UI/UX, are owned by the API/deployment/client areas — the database area covers only the facets a data layer actually owns and names the delegate. The 13 covered types each cite a runnable skeleton; none is asserted without a concrete test location. The Postgres-path caveats (`ATM-DB-001` placeholder rewrite, `ATM-DB-002` `CONCURRENTLY`) are the paired-mutation gates that make a green run *prove real behaviour* rather than a SQLite-only false pass.

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